

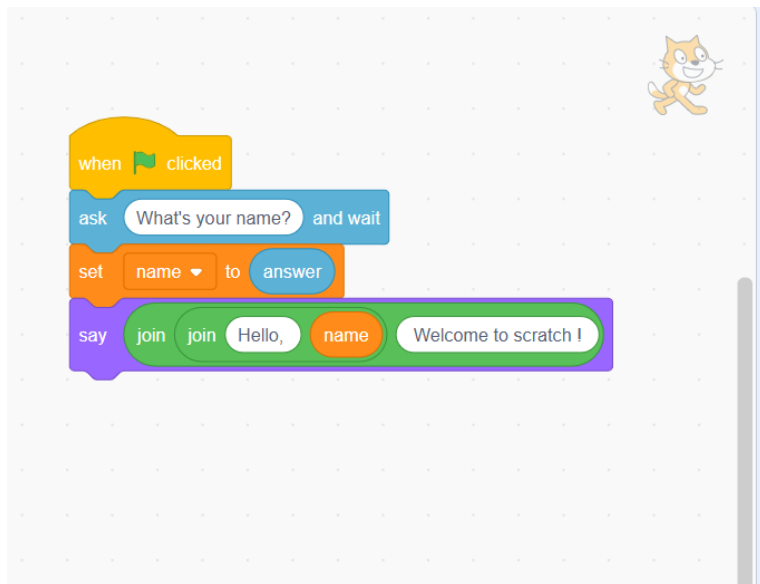
Objectives of the lab:

1. Learn to take input from users and display output in Scratch.
2. Understand and use variables to store and manipulate data.
3. Perform arithmetic operations using Operator blocks.
4. Use relational operators to enable decision-making.
5. Apply decision and loop blocks to control program flow.

Task 1: Personalized Greeting

Steps:

1. Start the program with when green flag clicked.
2. Ask the user for their name and store it in a variable name.
3. Display a personalized greeting using say (join [Hello,] (name)) [Welcome to Scratch!].

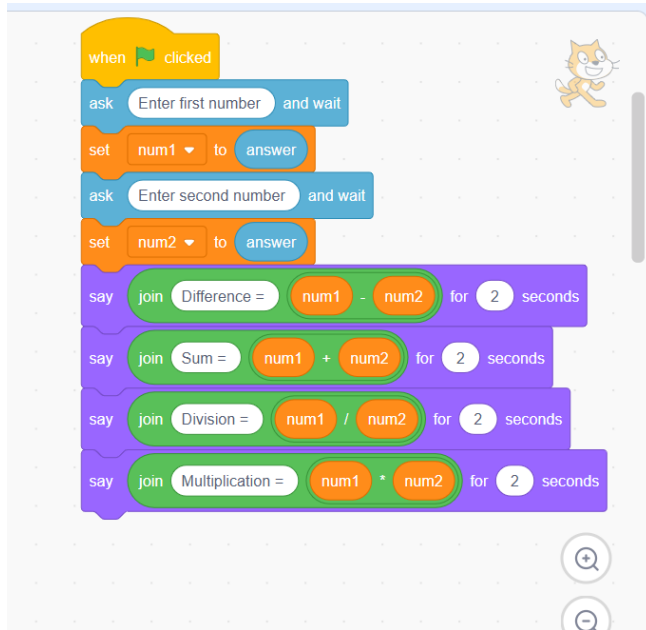


Task 2: Simple Calculator

Steps:

1. Start with when green flag clicked.
2. Ask the user for two numbers and store them in variables num1 and num2.

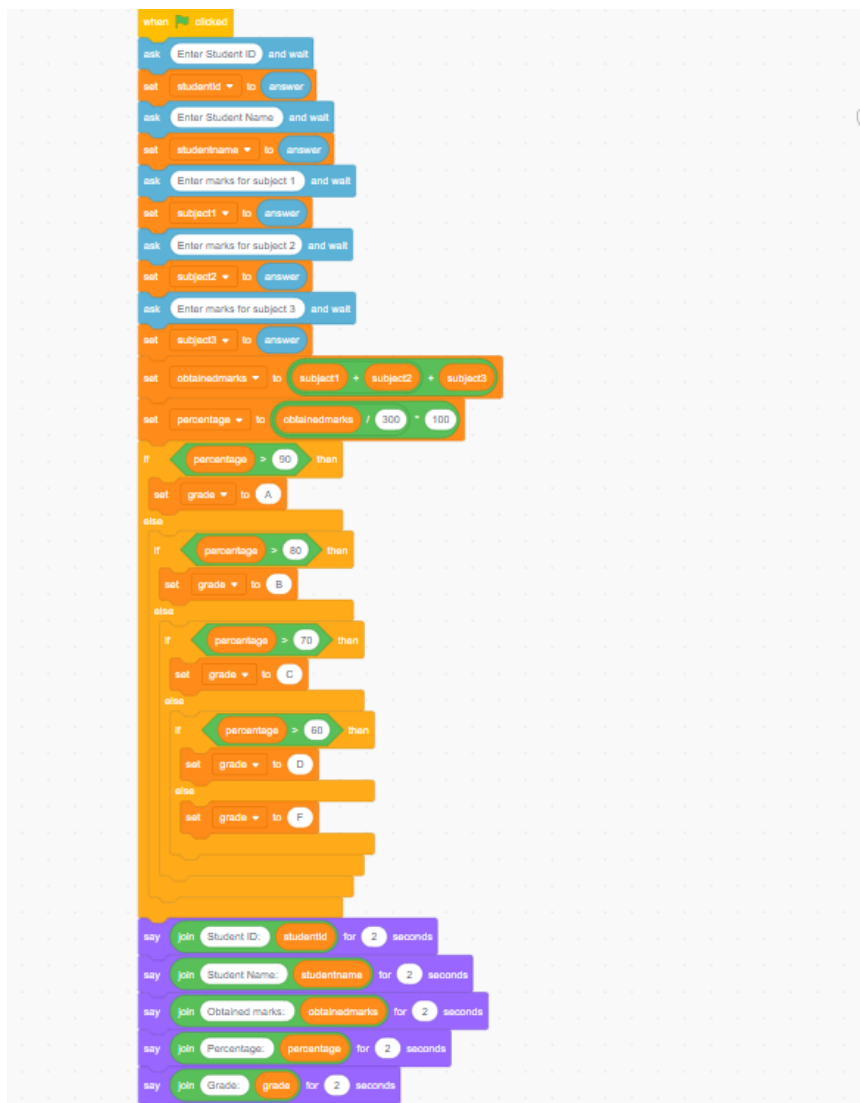
3. Use operators to calculate sum, difference, product, and quotient.
4. Display each result using say ... for seconds to make each visible.



Task 3: Student Details Record System

Steps:

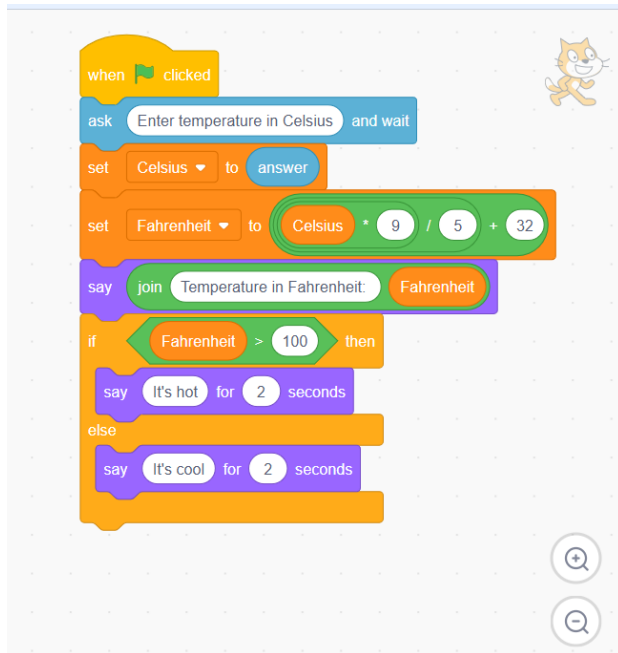
1. Start with when green flag clicked.
2. Ask the user for student ID, Name, and marks for three subjects; store them in variables studentID, studentname, subject1, subject2, subject3.
3. Calculate obtainedmarks = subject1 + subject2 + subject3.
4. Calculate percentage = (obtainedmarks / 300) * 100.
5. Assign a grade based on percentage using decision blocks (if ... else).
6. Display the student's details, obtained marks, percentage, and grade.



Task 4: Temperature Converter with Feedback

Steps:

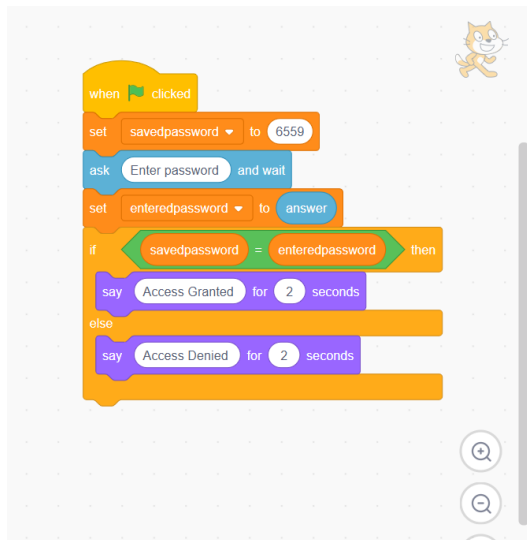
1. Start with when green flag clicked.
2. Ask the user for temperature in Celsius and store it in celsius.
3. Convert to Fahrenheit: $\text{fahrenheit} = (\text{celsius} * 9 / 5) + 32$.
4. Display Fahrenheit value.
5. Use a conditional block to display feedback: "It's hot!" if Fahrenheit > 100, else "It's cool!".



Task 5: Password Checker

Steps:

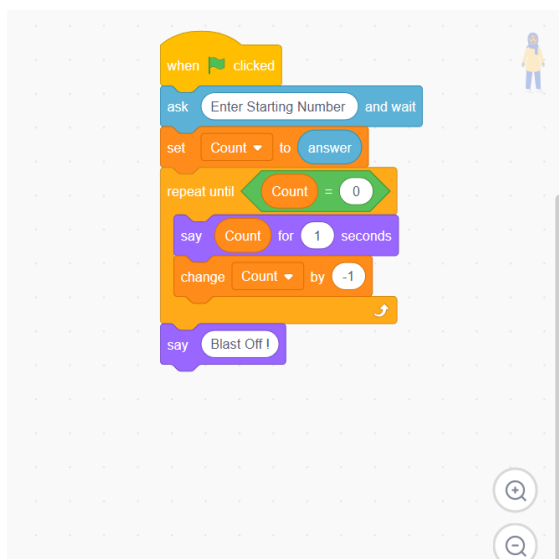
1. Start with when green flag clicked.
2. Store the correct password in storedPassword.
3. Ask the user to enter a password and store it in enteredPassword.
4. Compare using if ... else.
5. Display "Access Granted" if correct, else "Access Denied".



Task 6: Countdown Timer

Steps:

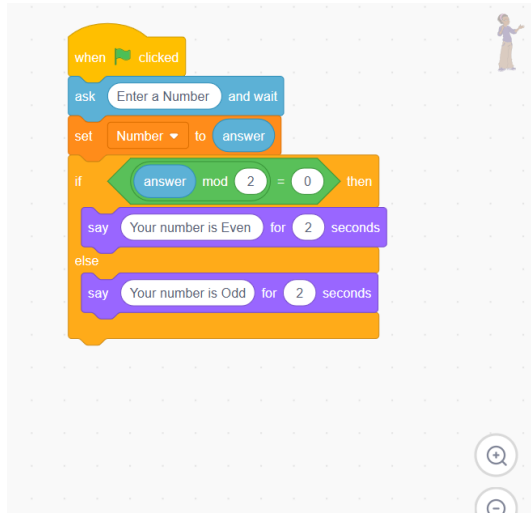
1. Start with when green flag clicked.
2. Ask the user for starting number and store in count.
3. Use a repeat until (count = 0) loop:
 - Display count using say ... for 1 second.
 - Decrease count by 1 each time.
4. After loop ends, display "Blast Off!".



Task 7: Even / Odd Number Identifier

Steps:

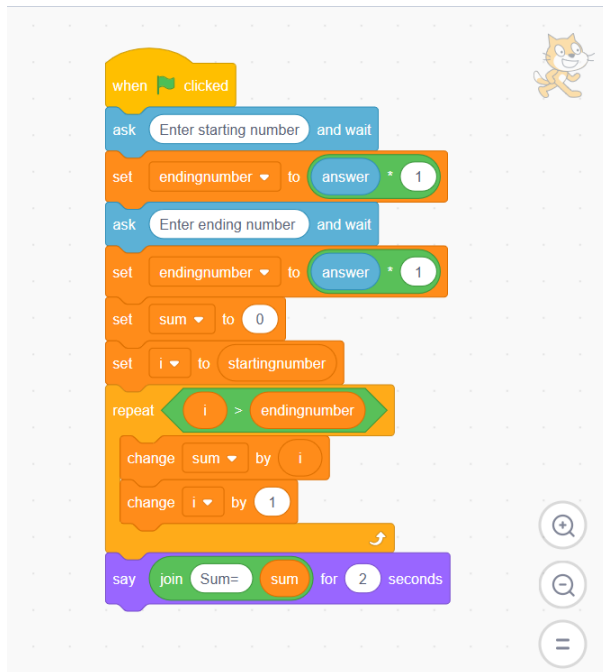
1. Start with when green flag clicked.
2. Ask the user for a number and store in number.
3. Use modulo operator (number mod 2) to check remainder.
4. Use if ... else to display “Your number is even” or “Your number is odd”.



Task 8: Sum of a User-Defined Range

Steps:

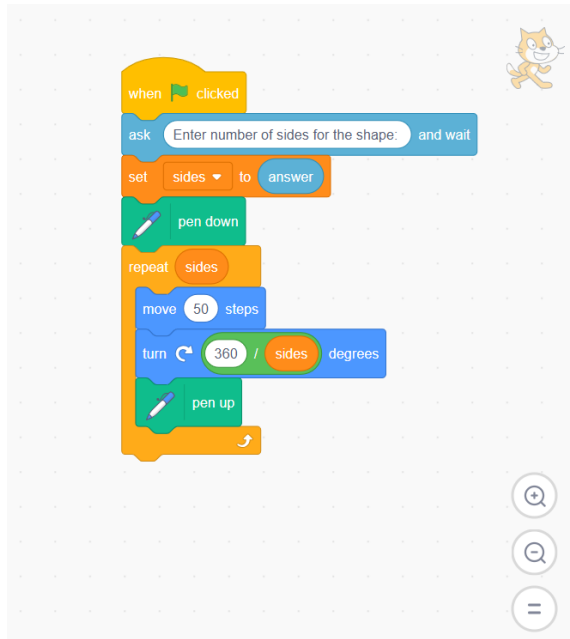
1. Start with when green flag clicked.
2. Ask user for starting and ending numbers, store in startnumber and endnumber.
3. Initialize sum = 0 and i = startnumber.
4. Use repeat until (i > endnumber) loop:
 - Add i to sum.
 - Increase i by 1.
5. Display total sum after loop ends.



Task 9: Interactive Shape Drawer

Steps:

1. Start with when green flag clicked.
2. Ask user for number of sides and store in sides.
3. Use Pen Extension to draw.
4. Put pen down, then repeat (sides) loop:
 - Move 50 steps.
 - Turn $(360 / \text{sides})$ degrees.
5. Lift pen after drawing with pen up.



Task 10: Number Guessing Game

Steps:

1. Start with when green flag clicked.
2. Generate random number between 1-10 and store in secretnumber.
3. Use repeat until (guess = secretnumber) loop:
 - Ask user for guess, store in guess.
 - If guess > secretnumber → say "Too high".
 - If guess < secretnumber → say "Too low".
 - Else → say "Correct!".

